

FAHIM ANZUM

Ph.D. Candidate, Computer Science · Applied ML & NLP Researcher · Explainable & Trustworthy AI

✉ fahim.anzum@ucalgary.ca · 🌐 anzumbivor.github.io · 📧 Google Scholar · 📄 anzumbivor · 📧 anzumbivor

Summary

Computer Science Ph.D. candidate specializing in trustworthy, explainable, and human-centered AI, with a focus on modeling linguistic, affective, and psychological signals in human- and LLM-generated social media discourse. My research combines NLP, transformer-based and hybrid deep learning, affective computing, and social media analytics to develop interpretable AI systems for health, misinformation detection, and human-AI interaction. I have **6+ years** of applied machine learning and NLP research experience and **16 peer-reviewed publications** in venues including **TMLR**, **IEEE Access**, **IEEE ICHMS**, and **Springer Nature**, supported by competitive funding and interdisciplinary collaborations across health, public policy, biomedical research, and industry.

Education

- Ph.D., Computer Science** University of Calgary, Calgary, Alberta Expected Aug 2026
Committee: Marina L. Gavrilova, Usman Alim, Michael Ulliyot, Mario Costa Sousa, Ehud Sharlin.
Thesis: *Trajectory-Based Modeling of Language and Emotion in Human and LLM-Generated Social Media Text for Trustworthy and Explainable Decision-Making.*
- M.Sc., Computer Science** University of Calgary, Calgary, Alberta Aug 2021
Advisors: Mario Costa Sousa, Usman Alim.
Thesis: *Exploring Convolutional Neural Networks and Transfer Learning for Oil Sands Drill Core Image Analysis.*
- B.Sc., Computer Science and Engineering** Military Institute of Science and Technology (MIST), Bangladesh Jan 2018
Advisors: Muhammad Nazrul Islam, Wali Mohammad Abdullah.

Technical Expertise

Programming & ML Frameworks: Python, PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, Java.

Transformers & LLM Experimentation: BERT, RoBERTa, DistilBERT, Mistral-7B-Instruct-v0.3, Meta-Llama-3.1-8B-Instruct, ChatGPT API, open-weight LLMs, API-based prompting, transformer fine-tuning.

Explainability, Safety & Evaluation: LIME, SHAP, Integrated Gradients, attention-based attribution, XAI faithfulness evaluation, MoRF/deletion analysis, comprehensiveness, sufficiency, bias and fairness analysis, artifact-controlled evaluation, statistical significance testing.

Modeling & Analytics: Deep learning, NLP, temporal and sequence modeling, time-series forecasting, statistical modeling, feature engineering, topic modeling, opinion mining, affective and behavioral modeling, social computing.

Research Engineering & Systems: End-to-end ML pipelines, large-scale text preprocessing, feature extraction, model benchmarking, ablation studies, reproducible experimentation, Linux, Bash scripting, Git.

Publications

Submitted and Under Review Manuscripts

- [UR2]** Fahim Anzum, A S M Hossain Bari, and Marina L. Gavrilova. “xEmoRec: A Unified Architecture for Explainable Emotion Recognition toward Trustworthy Human-Machine Systems.” Submitted to *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, 2026. [Code]
- [UR1]** Fahim Anzum and Marina L. Gavrilova. “EmoBlend Fusion: Towards Trustworthy Human-Machine Teaming through Enhanced Emotion Detection in Microblogs.” Under review in *IEEE Transactions on Human-Machine Systems*, 2026.

Manuscript in Preparation

- [IP1]** Fahim Anzum and Marina L. Gavrilova. “Trajectory-Based Modeling of Affective and Linguistic Dynamics for Human- vs. LLM-Generated Misinformation Detection.” In preparation for *WinLP @ EMNLP*, 2026.

Published Peer-Reviewed Conference, Journal, and Book Chapter Papers

- [P16]** Lily Dey, Fahim Anzum, Ulises Charles-Rodriguez, A S M Hossain Bari, Jean-Christophe Boucher, Aleem Bharwani, and Marina L. Gavrilova. “Exploring Public Trust Through LLM-Driven Opinion Mining.” In *International Conference on Computer Information Systems and Industrial Management (CISIM)*, pp. 59–74, Springer Nature, 2025. [PDF] [Code]
- [P15]** Pedro Paiva, Farzaneh Dehghani, Fahim Anzum, Mansi Singhal, Ayah Metwali, Marina Gavrilova, and Mariana

Bento. "FairHealthGrid: A Systematic Framework for Evaluating Bias Mitigation Strategies in Healthcare Machine Learning." In *31st Americas Conference on Information Systems (AMCIS)*, Vol. 46, 2025.

- [P14] Farzaneh Dehghani, Mahsa Dibaji, **Fahim Anzum**, Lily Dey, Alican Basdemir, Sayeh Bayat, Jean-Christophe Boucher, Steve Drew, Sarah Elaine Eaton, Richard Frayne, Gouri Ginde, Ashley Harris, Yani Ioannou, Catherine Lebel, John Lysack, Leslie Salgado Arzuaga, Emma Stanley, Roberto Souza, Ronnie de Souza Santos, Lana Wells, Tyler Williamson, Matthias Wilms, Zaman Wahid, Mark Ungrin, Marina Gavrilova, and Mariana Bento. "Trustworthy and Responsible AI for Human-Centric Autonomous Decision-Making Systems." *Transactions on Machine Learning Research (TMLR)*, 2025.
- [P13] **Fahim Anzum** and Marina L. Gavrilova. "EmoBlend Fusion: Leveraging Handcrafted and Deep Features for Emotion Detection." In *IEEE International Conference on Human-Machine Systems (ICHMS)*, pp. 1–6, 2024. [\[DOI\]](#) [\[Best Student Paper\]](#)
- [P12] **Fahim Anzum**, Ashratuz Zavin Asha, Lily Dey, Artemy Gavrilov, Fariha Iffath, Abu Quwsar Ohi, Liam Pond, Md Shopon, and Marina L. Gavrilova. "A Comprehensive Review of Trustworthy, Ethical, and Explainable Computer Vision Advancements in Online Social Media." In *Global Perspectives on the Applications of Computer Vision in Cybersecurity*, pp. 1–46, 2024.
- [P11] **Fahim Anzum** and Marina L. Gavrilova. "Emotion Detection From Micro-Blogs Using Novel Input Representation." *IEEE Access*, Vol. 11, pp. 19512–19522, 2023. [\[DOI\]](#)
- [P10] Zaman Wahid, A S M Hossain Bari, **Fahim Anzum**, and Marina L. Gavrilova. "Human Micro-Expression: A Novel Social Behavioral Biometric for Person Identification." *IEEE Access*, Vol. 11, pp. 57481–57493, 2023.
- [P9] **Fahim Anzum**, Hamidreza Hamdi, Usman Alim, and Mario Costa Sousa. "Exploring Convolutional Neural Networks and Machine Learning for Oil Sands Drill Core Image Analysis." In *EAGE Digitalization Conference and Exhibition*, pp. 1–5, 2023.
- [P8] Marina L. Gavrilova, **Fahim Anzum**, A S M Hossain Bari, Yajurv Bhatia, Fariha Iffath, Quwsar Ohi, Md Shopon, and Zaman Wahid. "A Multifaceted Role of Biometrics in Online Security, Privacy, and Trustworthy Decision Making." In *Breakthroughs in Digital Biometrics and Forensics*, Springer, pp. 303–324, 2022.
- [P7] **Fahim Anzum**, Ashratuz Zavin Asha, and Marina L. Gavrilova. "Biases, Fairness, and Implications of Using AI in Social Media Data Mining." In *IEEE International Conference on Cyberworlds (CW)*, pp. 251–254, 2022.
- [P6] Kaitlin De Chastelain Finnigan, **Fahim Anzum**, Jon Rokne, and Marina L. Gavrilova. "Weighted Lexicon-Based Sentiment Analysis for Women Career Traits in Information Technology." In *IEEE International Conference on Cognitive Informatics & Cognitive Computing (ICCI*CC)*, pp. 91–98, 2022. [\[Best Paper Award\]](#)
- [P5] Ashratuz Zavin Asha, **Fahim Anzum**, Patrick Finn, Ehud Sharlin, and Mario Costa Sousa. "Designing External Automotive Displays: VR Prototypes and Analysis." In *ACM International Conference on Automotive User Interfaces and Interactive Vehicular Applications (AutoUI)*, pp. 74–82, 2020.
- [P4] Ashratuz Zavin Asha, **Fahim Anzum**, S. M. Faisal Rahman, Muhammad Nazrul Islam, and Mehreen Hoque. "Towards Developing an Intelligent Fire Exit Guidance System Using Informed Search Technique." In *International Conference of Computer and Information Technology (ICCIT)*, pp. 1–6, 2018.
- [P3] Adnan Sharif, **Fahim Anzum**, Ashratuz Zavin Asha, Sayma Alam Suha, Anika Ibnat, and Muhammad Nazrul Islam. "Exploring the Opportunities and Challenges of Adopting Augmented Reality in Education in a Developing Country." In *IEEE International Conference on Advanced Learning Technologies (ICALT)*, pp. 364–366, 2018.
- [P2] Fahim Ahmed, **Fahim Anzum**, Muhammad Nazrul Islam, Wali Mohammad Abdullah, Sazid Al Ahsan, and Moneruzzaman Rana. "A New Algorithm to Compute Single Source Shortest Path in a Real Edge Weighted Graph to Optimize Time Complexity." In *IEEE International Conference on Computer and Information Science (ICIS)*, pp. 185–191, 2018.
- [P1] **Fahim Anzum**, Fahim Ahmed, M. Shariful Azim, Mosaddek Hossain, Shifat Zaman, Farhan Hasib, and Sazid Al Ahsan. "Smart Self Position Aligning Chair for a Modern Conference Room." In *IEEE International Conference on Computer and Information Science (ICIS)*, pp. 263–268, 2018.

Research Experience

Graduate Assistant – Research / Applied Machine Learning Researcher *Biometric Technologies* Sep 2021 – Present
Lab, University of Calgary, Canada

- Leading research on **human- vs. LLM-generated misinformation detection**, including the curation of controlled fake-news datasets generated using Mistral, LLaMA, Vicuna, and ChatGPT/API-based prompting; developing trajectory-based deep learning models that leverage linguistic, affective, and discourse-level signals for robust and generalizable detection beyond surface-level textual artifacts.
- Designed **SSEL**, a novel feature representation for emotion detection from social media microblogs, integrating stylistic, sentiment, and linguistic features with genetic algorithm-based feature optimization to improve emotion classification while preserving interpretability. [\[IEEE Access'23\]](#)
- Spearheaded the development of **EmoBlend Fusion**, an interpretable hybrid model that combines handcrafted SSEL

- features with transformer-derived deep representations for enhanced emotion recognition and trustworthy human-AI interaction; the initial conference paper received **Best Student Paper** recognition at IEEE ICHMS 2024. [\[IEEE ICHMS'24\]](#)
- Designed **xEmoRec**, a unified explainable emotion recognition model integrating RoBERTa-CNN-BiLSTM with CLS-query multi-head attention, achieving **93.88% accuracy** and **0.93 macro-F1** on a **416K+ microblog emotion dataset**. [\[GitHub\]](#)
 - Extended xEmoRec with a systematic **XAI faithfulness evaluation framework**, comparing attention-based attribution and Integrated Gradients through deletion/MoRF curves, deletion AUC, comprehensiveness, sufficiency, and statistical significance testing to evaluate whether token-level explanations reflect model decision behavior.
 - Applied **NLP, sentiment analysis, social media mining, and BERTopic modeling** to analyze large-scale public discourse on trust in Canadian healthcare systems, extracting behavioral and affective signals from unstructured Reddit discussions and translating findings into policy- and governance-relevant insights for healthcare stakeholders. [\[CISIM 2025\]](#)
 - Led transdisciplinary collaborations with **Alberta Health Services**, the **Cumming School of Medicine**, the **Department of Political Science**, the **School of Public Policy**, and public health collaborators, bridging AI research with applications in healthcare, governance, social innovation, and public policy.
 - Built end-to-end **Python/PyTorch and Hugging Face pipelines** across multiple projects for large-scale data scraping, preprocessing, feature engineering, model development, model training, benchmarking, ablation studies, artifact-controlled evaluation, and reproducible experimentation.
 - Mentored and supervised graduate and undergraduate students in research design, technical implementation, experimental validation, and academic writing, with mentees contributing to peer-reviewed publications and awards.
 - Contributed to grant writing, proposal development, academic workshops, and seminars aligned with institutional and national priorities in trustworthy AI, ethical social media data mining, and AI for social innovation.

Industry Research Experience

Research Scientist Intern, Mitacs-Accelerate Program *Suncor Energy Inc., Calgary, Canada* Sep 2019 – Aug 2021

- Conducted applied machine learning research under the Mitacs-Accelerate Graduate Research Program, developing computer vision and machine learning models to advance data-driven geoscience analytics for oil sands drill core imagery.
- Pioneered a **hybrid ML-CNN model** for facies and mean grain-size prediction from drill core images, achieving an approximately **20% performance improvement** over Suncor's existing system.
- Built reproducible end-to-end pipelines for image preprocessing, feature extraction, model training, cross-validation, and evaluation, with emphasis on integration into downstream geoscience analytics workflows.
- Supervised a team of **five geologists** in data collection, annotation, and preprocessing of proprietary oil sands drill core imagery for model development and validation.
- Collaborated with a multidisciplinary core team of ML scientists, software engineers, geologists, geoscientists, and engineers to evaluate model generalizability, interpretability, operational constraints, production reliability, and integration feasibility within Suncor's analytics pipeline.
- Presented research outcomes to Suncor's data science and R&D leadership, contributing to the company's digital transformation and intelligent analytics initiatives.

Teaching Experience

Sessional Instructor, Department of Computer Science *University of Calgary, Canada* Sep 2019 – Present

Course ownership · Undergraduate instruction · Enrollment: 160+

- Independently designed and delivered **CPSC 231: Introduction to Computer Science for Computer Science Majors I**, an introductory Python programming course emphasizing algorithmic thinking, problem decomposition, and code readability.
- Developed course outlines, lecture materials, assignments, labs, examinations, and assessment rubrics, integrating live coding, interactive demonstrations, and in-class activities to support diverse learning styles.
- Implemented formative and summative assessments to reinforce conceptual understanding and support progressive skill development in programming.
- Mentored students through structured office hours and individualized feedback, fostering an inclusive, supportive, and engaging learning environment.
- Supervised and coordinated teaching assistants, ensuring consistency in grading practices, tutorial delivery, and instructional quality across course components.
- Organized a guest lecture and industry networking session to connect course content with real-world applications and professional pathways.

Graduate Teaching Assistant / Head Teaching Assistant *Department of Computer Science, University of Calgary, Canada* Sep 2019 – Present

Undergraduate and graduate instruction · Enrollment: 400+

- Led tutorials and lab sessions for undergraduate and graduate courses, including **CPSC 217: Introduction to Computer Science for Multidisciplinary Studies I**, **CPSC 319: Data Structures, Algorithms, and Their Application**, **DATA 604: Working with Data at Scale**, and **DATA 201: Thinking with Data**.
- Designed tutorial materials, supplementary resources, and grading rubrics aligned with course learning objectives.
- Served as Head TA in multiple courses, coordinating TA teams, managing grading workflows, and maintaining instructional consistency across large course sections.
- Provided academic mentoring and one-on-one student support during weekly office hours.

Lecturer, Department of Computer Science and Engineering *United International University and Primeasia University, Bangladesh* Mar 2018 – Aug 2019

Undergraduate instruction · Enrollment: 800+

- Independently taught undergraduate theory and laboratory courses, including **Computer Graphics**, **Graph Theory**, **Software Engineering**, **Discrete Mathematics**, **Computer Networks**, and **Introduction to Computers and Computer Fundamentals Lab**.
- Designed course outlines, lectures, and assessments using project-based learning, presentations, quizzes, midterms, final exams, and lab-based evaluation.
- Supervised undergraduate capstone projects, mentoring students in research design, technical implementation, and academic writing.
- Completed faculty development and mentorship training programs to strengthen pedagogical practice and instructional effectiveness.

Scholarships, Awards, and Honors

- **Open Doctoral Scholarship**, Faculty of Graduate Studies, University of Calgary, CA\$15,000, January 2026.
- **Best Student Paper Award (Top 3)**, IEEE International Conference on Human-Machine Systems (ICHMS), 2024.
- **Best Paper Award**, IEEE International Conference on Cognitive Informatics & Cognitive Computing (ICCI*CC), 2022.
- **Alberta Innovates Scholarship**, Government of Alberta, CA\$124,000 over four years, 2021.
- **Eyes High International Doctoral Recruitment Scholarship**, University of Calgary, CA\$60,000, 2021.
- **Mitacs-Accelerate Graduate Research Internship**, University of Calgary, CA\$50,000, 2019.
- **Graduate Research Award**, Department of Computer Science, University of Calgary, CA\$11,000, 2020.
- **International Graduate Tuition Award**, University of Calgary, CA\$12,000 over four years, 2021.
- **International Graduate Recruitment Award**, University of Calgary, CA\$2,000, 2019.
- **Visa Differential Scholarship**, University of Calgary, CA\$4,126.17, 2019.
- **Commandant's List Award**, MIST, awarded for maintaining GPA above 3.80, 2015 and 2016.
- **National Merit Scholarship**, Government of Bangladesh, 2008, 2010, and 2012.
- **National Secondary School Merit Scholarship**, Government of Bangladesh, 2010.
- **National High School Merit Scholarship**, Government of Bangladesh, 2008.

Academic Service

Academic Hiring Committee

- Served as graduate student representative for the tenure-track research-stream faculty hiring committee in the Department of Computer Science, University of Calgary, Spring 2024.

Reviewer: Conferences, Journals, and Books

- ACM KDD Workshop on Generative AI for Recommender Systems and Personalization, papers, 2024.
- ACM CHI Late-Breaking Works, 2024.
- IEEE Access, regular papers, 2022, 2023, 2024.
- Springer *The Visual Computer*, regular papers, 2024.
- Springer *Transactions on Computational Science*, papers, 2022, 2023, 2024.
- IEEE ICCSA, papers, 2022, 2023.
- Springer Nature book proposal on *Biases and Ethics in Social Intelligence*.

Conference Program Committee and Volunteering

- Technical Program Committee Member, 12th International Symposium on Applied Computing for Software and Systems (ACSS), Kolkata, India, 2025.

- Organizing and Logistics Sub-Committee Member, IEEE 21st International Conference on Computer and Information Technology (ICCIT), Dhaka, Bangladesh, 2018.
- Student Volunteer, ACM DIS 2023, Pittsburgh, USA.

University Service: University of Calgary

- Academic Event Committee Member, Graduate Students' Association (GSA), Sep 2022 – Apr 2023.
- Vice-President Communications, Computer Science Graduate Society (CSGS), Apr 2023 – Apr 2024.
- Judge, CalgaryHacks Competitive Programming Contest, Feb 2022.

Mentoring and Supervision

Mentored students in research design, technical implementation, academic writing, and professional development, with mentees contributing to peer-reviewed publications, conference submissions, and student research awards.

- **Lily Dey**, Master's student, Department of Computer Science, University of Calgary, Jan 2024 – Present.
- **Wamika Jha**, Master's student, Department of Computer Science, University of Calgary, May 2024.
- **Zaman Wahid**, Master's student, Department of Computer Science, University of Calgary, Jan 2022 – Jun 2023.
- **Kaitlin De Chastelain Finnigan**, Bachelor's student, Department of Computer Science, University of Calgary, Jun 2022 – Dec 2022.

Selected Invited Talks, Seminars, and Posters

- **Guest Lecture**, United International University, Bangladesh, Jan 2023. Talk: *Ethics in Online Social Media Data Mining*.
- **Poster Presentation**, University of Calgary, Canada, Sep 2022. Poster: *AI in Social Media Data Mining: Viewing Through the Lens of Ethical Aspects*.
- **Guest Lecture**, Bournemouth University, England, Sep 2022. Talk: *Deep Learning in Social Media Opinion Mining*.
- **Guest Lecture**, DATA 501 – Data Science Capstone, University of Calgary, Canada, Jul 2022. Talk: *Introduction to Deep Learning and Convolutional Neural Networks*.
- **Invited Talk**, Visualization and Graphics Group, University of Calgary, Canada, Jul 2022. Talk: *Drill Core Image Classification using Convolutional Neural Networks*.
- **M.Sc. Graduate Seminar**, University of Calgary, Canada, Jul 2021. Talk: *Exploring Transfer Learning and Convolutional Neural Networks for Predicting Mean Grain Size from Oil Sands Drill Core Images*.